

Multivariable Calculus

Limits · Continuity · Introduction to Derivatives

Week	01	Lecture	01 – 02
Programme	BS Computing	Semester	3rd
Reference	Thomas' Calculus		



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Learning Objectives

By the end of this lecture, you will be able to:

- Distinguish between single-variable and multivariable functions.
- Evaluate limits of functions of one, two, and three variables.
- Apply the three-part definition of continuity, including left-hand and right-hand limits.
- Test multivariable continuity along different paths and recognize path-dependence.
- Understand the derivative as an instantaneous rate of change, as a bridge to partial derivatives.

1

Introduction — Functions of One and Several Variables

Calculus, at its core, is the mathematics of **change**. Single-variable calculus studies how a quantity changes with respect to one independent variable; multivariable calculus extends these same ideas — limits, continuity, and derivatives — to functions that depend on **two or more** independent variables simultaneously. This is essential for computing applications such as machine learning loss surfaces, 3D graphics, physics simulations, and optimization algorithms, all of which operate in many-dimensional spaces.

Single-Variable Function

$f(x) = 2x \longrightarrow$ dependent variable

Here x is the **independent** variable and $f(x)$ is the **dependent** variable — it depends on the single input x .

Multivariable Function

$f(x, y) = 2x + 7y$

Here f depends on **two** independent variables, x and y . The output is a surface in 3-dimensional space rather than a curve in 2 dimensions.

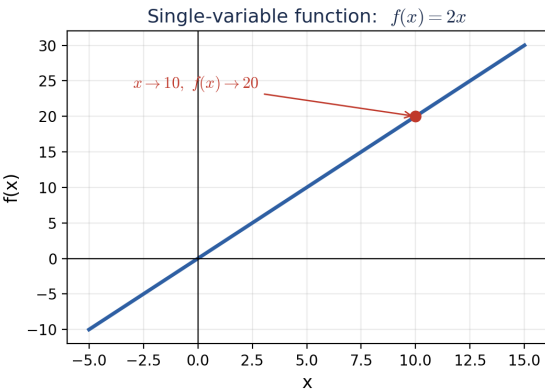


Fig 1.1 — A single-variable function $f(x) = 2x$ is a curve in the xy -plane.

Table 1.1 — Quick Comparison

Property	Single-Variable	Multivariable
Independent variables	One (x)	Two or more ($x, y, z, \dots$)
Typical notation	$f(x)$	$f(x, y), f(x, y, z)$
Graph	A curve in the plane	A surface (or hypersurface) in space
Approach directions at a point	2 — from the left or right	Infinitely many — every direction in the plane
Domain	A subset of the real line	A subset of the plane (or higher-dim. space)

■ Why This Matters for Computing

Every idea in this lecture — limits, continuity, derivatives — has a single-variable version you already know from earlier calculus courses. The multivariable version is not a different theory; it is the same definition applied where the input has more than one component, and the extra directions are exactly what makes it richer (and occasionally trickier).

2

Limits

A limit describes the value a function *approaches* as its input approaches some point — the input itself need not ever reach that point, and the function need not even be defined there. Limits are the foundation on which continuity and derivatives are built.

2.1 Single-Variable Limits

Definition

The notation below is read “the limit of $f(x)$ as x approaches 10”:

$$\lim_{x \rightarrow 10} f(x) = 20$$

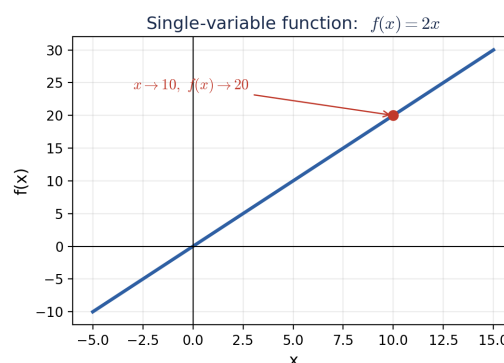


Fig 2.1 — As $x \rightarrow 10$, $f(x) = 2x$ approaches 20 (marked point).

2.2 Multivariable Limits

For a function of two variables, the point (x, y) approaches a target point from every direction in the plane at once — not just from the left or right as in one dimension. The limit exists only if $f(x, y)$ approaches the **same** value no matter which path (x, y) takes toward that point.

Definition — Two and Three Variables

$$\lim_{(x, y) \rightarrow (1, 2)} f(x, y) = \lim_{(x, y) \rightarrow (1, 2)} (2x + 3y)$$

$$\lim_{(x, y, z) \rightarrow (1, 2, 3)} f(x, y, z) = \lim_{(x, y, z) \rightarrow (1, 2, 3)} (2x + 3y + 4z)$$

Multivariable function: $f(x, y) = 2x + 3y$

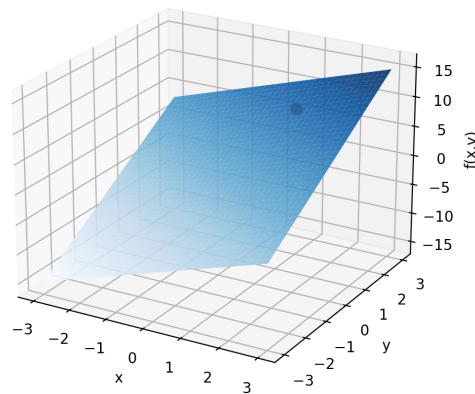


Fig 2.2 — The plane $f(x, y) = 2x + 3y$. As $(x, y) \rightarrow (1, 2)$, the surface value approaches the marked point.

2.3 Worked Examples

Example 1 — Rational Multivariable Limit

$$\lim_{(x, y) \rightarrow (0, 1)} \frac{x - xy + 3}{x^2 y + 5xy - y^3}$$

Substitute $(x, y) = (0, 1)$ directly, since the denominator does not vanish there:

$$= \frac{0 - (0)(1) + 3}{(0)^2(1) + 5(0)(1) - (1)^3} = \frac{3}{-1} = -3$$

Example 2 — Limit of a Radical (Distance Function)

$$\lim_{(x, y) \rightarrow (3, -4)} \sqrt{x^2 + y^2}$$

This is the Euclidean distance function – substitute directly:

$$= \sqrt{3^2 + (-4)^2} = \sqrt{9 + 16} = \sqrt{25} = 5$$

■ Key Idea

Direct substitution works whenever the resulting expression is defined (no division by zero, no negative square root, etc.). When substitution gives an indeterminate form such as $0/0$, algebraic manipulation or path analysis (Section 3.3) is required.

■ Common Mistakes to Avoid

Forgetting to check the denominator. Always confirm the denominator is nonzero at the target point before substituting directly — otherwise the limit needs algebraic work (factoring, rationalizing) or is undefined.

Assuming one path is enough (multivariable case). A single matching path is evidence, not proof — see Section 3.3.

Mixing up LHL and RHL notation. The superscript on c indicates the *direction of approach* (c^- from below, c^+ from above), not an exponent.

3

Continuity of Functions

Informally, a function is continuous at a point if its graph can be drawn through that point without lifting the pen. Formally, three conditions must all hold.

3.1 The Three-Part Definition

A function f is continuous at $x = c$ if and only if:

1. $f(c)$ is defined — the functional value is defined
2. $\lim_{x \rightarrow c} f(x)$ exists — left-hand limit = right-hand limit
3. $f(c) = \lim_{x \rightarrow c} f(x)$ — the limit equals the functional value

Left-Hand Limit (LHL)

$$\text{LHL} = \lim_{x \rightarrow c^-} f(x)$$

Value $f(x)$ approaches as $x \rightarrow c$ from below ($x < c$).

Right-Hand Limit (RHL)

$$\text{RHL} = \lim_{x \rightarrow c^+} f(x)$$

Value $f(x)$ approaches as $x \rightarrow c$ from above ($x > c$).

Rule: if $\text{LHL} = \text{RHL}$, the (two-sided) limit exists — and if that limit also equals $f(c)$, the function is continuous at that point.

3.2 Single-Variable Continuity — Worked Examples

Example 1 — Discontinuity Test

$$f(x) = \begin{cases} 3x & x = 2 \\ 3x + 4 & x < 2 \\ 4x + 4 & x > 2 \end{cases}$$

Step 1 — Functional value:

$$f(2) = 3(2) = 6$$

Step 2 — One-sided limits:

$$\text{LHL} = \lim_{x \rightarrow 2^-} (3x + 4) = 3(2) + 4 = 10$$

$$\text{RHL} = \lim_{x \rightarrow 2^+} (4x + 4) = 4(2) + 4 = 12$$

Result:

$\text{LHL} \neq \text{RHL} \Rightarrow f(x)$ is discontinuous at $x = 2$

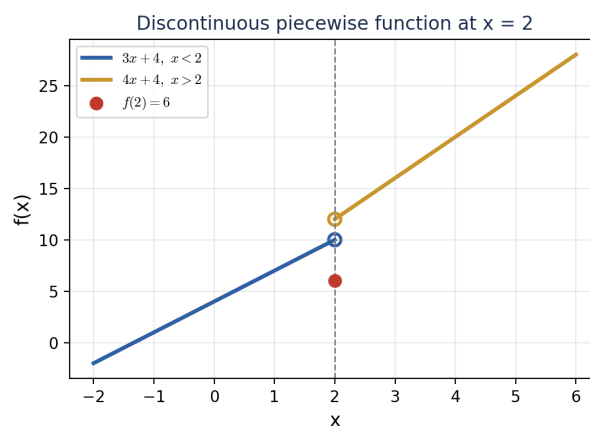


Fig 3.1 — The two branches meet different heights at $x = 2$ (open circles), so the function jumps: discontinuous.

Example 2 — Continuity Test

$$f(x) = \begin{cases} 2x + 5 & x \leq 2 \\ 4x + 1 & x > 2 \end{cases}$$

Step 1 — Functional value:

$$f(2) = 2(2) + 5 = 9$$

Step 2 — One-sided limits:

$$\text{LHL} = \lim_{x \rightarrow 2^-} (2x + 5) = 2(2) + 5 = 9$$

$$\text{RHL} = \lim_{x \rightarrow 2^+} (4x + 1) = 4(2) + 1 = 9$$

Result:

$f(2) = \text{LHL} = \text{RHL} = 9 \Rightarrow f(x)$ is continuous at $x = 2$

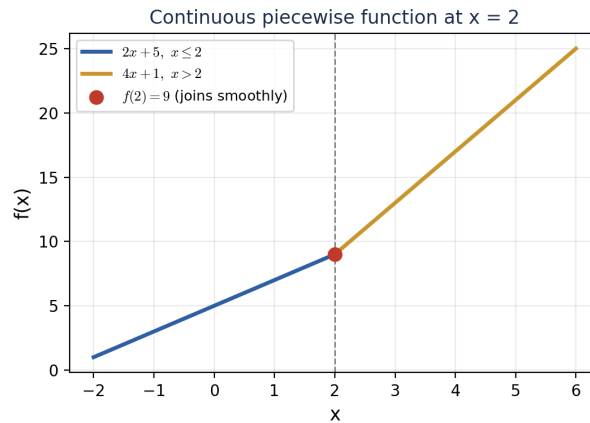


Fig 3.2 — The two branches meet at exactly the same point $(2, 9)$: the graph is unbroken — continuous.

3.3 Multivariable Continuity & Path Dependence

In two dimensions, (x, y) can approach a point from infinitely many directions — straight lines, parabolas, spirals, anything. For the limit to exist, **every** path must give the same value. Checking one path can suggest continuity, but never proves it on its own; checking two different paths that disagree is enough to prove the limit does **not** exist.

(x, y) can approach (a, b) from infinitely many directions

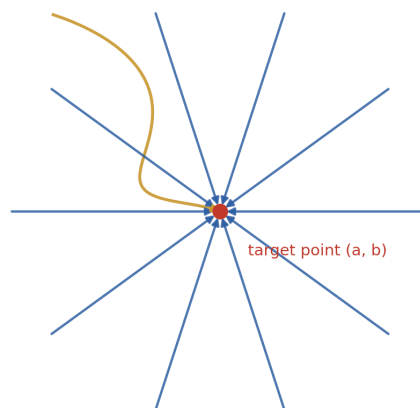


Fig 3.0 — In one dimension there are only two directions of approach; in the plane, (x, y) can approach a point along infinitely many straight or curved paths.

Example 3 — Testing Along a Path

$$f(x, y) = \begin{cases} \frac{x^2 y}{x^2 + y^2} & (x, y) \neq (0, 0) \\ 0 & (x, y) = (0, 0) \end{cases}$$

Step 1 — Along the path $y = x$:

$$\text{Along } y = x: \quad f(x, x) = \frac{x^2(x)}{x^2 + x^2} = \frac{x^3}{2x^2} = \frac{x}{2}$$

Step 2 — Functional value and limit:

$$f(0, 0) = 0 \qquad \lim_{(x, x) \rightarrow (0, 0)} \frac{x}{2} = 0$$

Result:

$$f(0, 0) = \lim_{(x, y) \rightarrow (0, 0)} f(x, y) = 0 \Rightarrow \text{continuous along } y=x$$

■ Rigor Check — Why One Path Isn't Enough

Testing a single path ($y = x$) only shows the function *behaves consistently along that path* — it does not by itself prove overall continuity. A rigorous proof bounds the function using the Squeeze Theorem, which holds for every possible path simultaneously:

$$\left| \frac{x^2 y}{x^2 + y^2} \right| \leq |y| \rightarrow 0 \quad (\text{Squeeze Theorem, holds for every path})$$

$$f(x, y) = \frac{x^2 y}{x^2 + y^2} \quad (\text{limit} = 0, \text{continuous at origin})$$

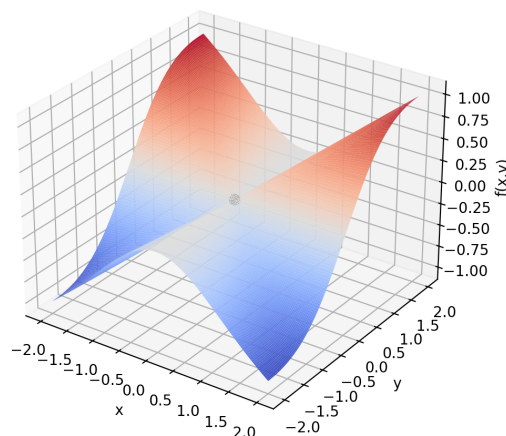


Fig 3.3 — The surface $f(x, y) = x^2 y / (x^2 + y^2)$ flattens smoothly to 0 at the origin from every direction.

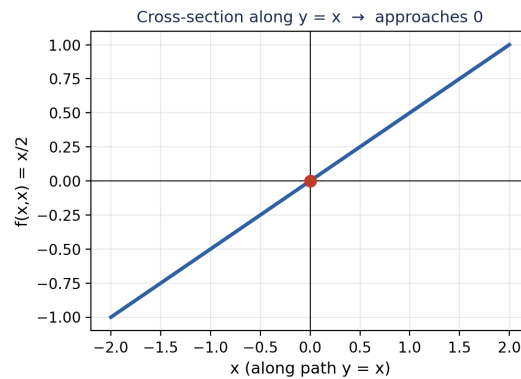


Fig 3.4 — Cross-section of the surface along the path $y = x$: $f(x,x) = x/2 \rightarrow 0$ as $x \rightarrow 0$.

Table 3.1 — Numerical Verification Along Several Paths

$f(x,y) = x^2y/(x^2+y^2)$ evaluated at shrinking distances from the origin, along three different paths — every column trends to 0:

x	y = x: $f(x,x) = x/2$	y = -x: $f(x,-x) = -x/2$	y = x ² : $f(x,x^2) = x^2/(1+x^2)$
1	0.5	-0.5	0.5
0.5	0.25	-0.25	0.2
0.1	0.05	-0.05	0.0099
0.01	0.005	-0.005	0.0001
$\rightarrow 0$	$\rightarrow 0$	$\rightarrow 0$	$\rightarrow 0$

This numerical pattern is consistent with (though, as noted above, does not by itself prove) the conclusion already established rigorously by the Squeeze Theorem: the limit is 0.

Example 4 — A Genuinely Discontinuous Case (Path Dependence)

Contrast the previous example with a classic case where different paths give **different** limits, so the overall limit does not exist:

$$g(x,y) = \frac{xy}{x^2 + y^2}, \quad (x,y) \neq (0,0)$$

$$\text{Along } y = 0: \quad g(x,0) = 0 \rightarrow 0$$

$$\text{Along } y = x: \quad g(x,x) = \frac{x^2}{2x^2} = \frac{1}{2} \rightarrow 0.5$$

Result:

$$0 \neq 0.5 \Rightarrow \lim_{(x,y) \rightarrow (0,0)} g(x,y) \text{ does not exist}$$

$$f(x, y) = \frac{xy}{x^2 + y^2} \quad (\text{limit depends on path — discontinuous})$$

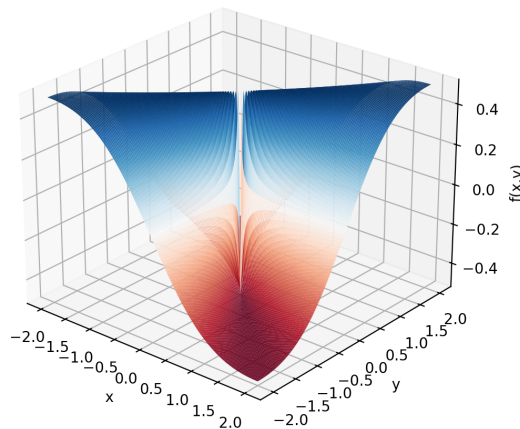


Fig 3.5 — $g(x, y) = xy/(x^2 + y^2)$. The surface spirals to different heights depending on the approach direction — there is no single value at the origin.

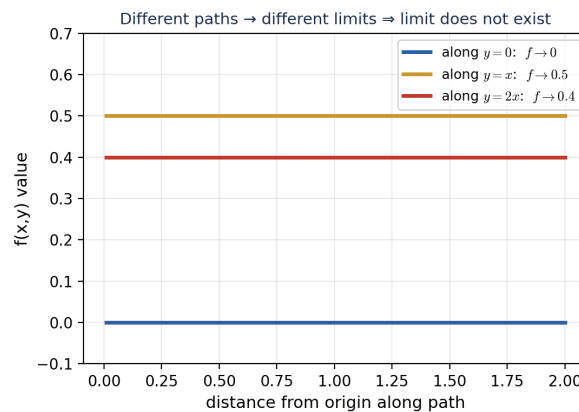


Fig 3.6 — Three different straight-line paths into the origin give three different limiting values: 0, 0.4, 0.5.

■ Why This Matters for Computing

This is the key qualitative difference between single- and multivariable continuity: in one dimension there are only two directions to check (left and right). In two or more dimensions, there are infinitely many, which is exactly why multivariable limits are structurally harder — and why tools like the Squeeze Theorem and polar coordinates become essential later in the course.

A Practical Checklist for Testing Multivariable Limits

- 1 **Try direct substitution first.** If the point is in the domain and nothing is undefined, you're done.
- 2 **Substitute along a family of straight lines** $y = mx$ (or $x = my$). If the result depends on the slope m , the limit does not exist — stop here.
- 3 **If straight lines all agree, try curved paths** such as $y = x^2$ or $y = x^3$, since some functions only reveal path-dependence along non-linear approaches.
- 4 **If every path tried agrees, attempt a rigorous bound** (Squeeze Theorem) or convert to polar coordinates ($x = r \cos\theta$, $y = r \sin\theta$) and check the limit as $r \rightarrow 0$ is independent of θ .
- 5 **Never conclude a limit exists from finitely many matching paths alone** — path-checking can only disprove existence conclusively; only a general bound proves it.

4

Introduction to Derivatives

Having established limits and continuity, we can now define the derivative — the tool that makes precise the idea of an **instantaneous rate of change**.

Rate of Change

$$\frac{df}{dx} \leftarrow \text{change in } f(x) \text{ (dependent)}$$

df is the change in the **dependent** variable and dx is the change in the **independent** variable. The derivative is the limit of this ratio as dx shrinks toward zero — it is an *instantaneous* rate of change, not an average one.

Formal Definition — The Difference Quotient

Limits make the idea of “instantaneous” precise. The derivative of f at x is defined as the limit, as h shrinks to zero, of the average rate of change over the interval $[x, x+h]$:

Definition

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

Worked Example — First Principles

Let $f(x) = x^2$. Find $f'(x)$ from first principles.

Step 1 — Expand $f(x+h)$:

$$f(x+h) = (x+h)^2 = x^2 + 2xh + h^2$$

Step 2 — Subtract $f(x)$:

$$f(x+h) - f(x) = (x^2 + 2xh + h^2) - x^2 = 2xh + h^2$$

Step 3 — Divide by h :

$$\frac{f(x+h) - f(x)}{h} = \frac{2xh + h^2}{h} = 2x + h$$

Step 4 — Take the limit as $h \rightarrow 0$:

$$f'(x) = \lim_{h \rightarrow 0} (2x + h) = 2x$$

For multivariable functions, this idea extends to **partial derivatives** — the rate of change of f with respect to one variable while holding every other variable fixed. These will be covered in detail in the next lecture:

$$\frac{\partial f}{\partial x}, \quad \frac{\partial f}{\partial y}$$

Multivariable function: $f(x, y) = 2x + 3y$

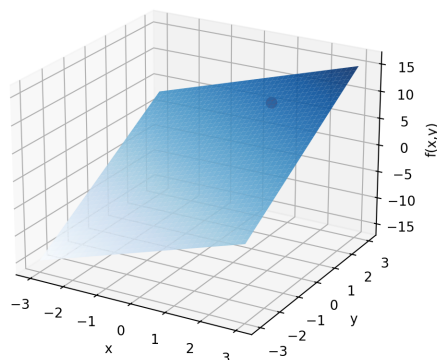


Fig 4.1 — On a multivariable surface, the rate of change along the x -direction and the y -direction can differ — this is the motivation for partial derivatives, covered next lecture.

5

Summary & Practice Problems

Key Takeaways

- A single-variable function depends on one input; a multivariable function depends on two or more, and its graph is a surface, not a curve.
- A limit is the value a function approaches as the input approaches a point — in 2D+ this must hold along every possible path.
- Continuity at $x = c$ requires: $f(c)$ defined, the limit at c exists (LHL = RHL), and the limit equals $f(c)$.
- One matching path suggests multivariable continuity but never proves it; one mismatched path is enough to disprove it.
- The derivative is the instantaneous rate of change — the limit of the difference quotient as the input increment shrinks to 0.

Practice Problems

- 1 Evaluate $\lim_{(x,y) \rightarrow (2,-1)} (3x^2y - 4y + 1)$.
- 2 Discuss the continuity of $f(x) = 5x+2$ for $x \leq 3$ and $2x+11$ for $x > 3$, at $x = 3$.
- 3 For $f(x,y) = xy^2/(x^2+y^4)$, test the limit at $(0,0)$ along the paths $y = x$ and $x = y^2$. What do you conclude?
- 4 Evaluate $\lim_{(x,y,z) \rightarrow (1,-2,3)} \sqrt{x^2+y^2+z^2}$.
- 5 Using the definition of the derivative as a limit, write (but do not yet solve) the difference quotient for $f(x) = x^2$ at a general point $x = a$.

Quick Reference Card

Concept	Formula / Rule
Single-variable limit	$\lim$ as $x \rightarrow c$ of $f(x)$
Multivariable limit	$\lim$ as $(x,y) \rightarrow (a,b)$ of $f(x,y)$ — must agree along every path
Continuity at $x = c$	$f(c)$ is defined AND the limit at c exists AND $\lim = f(c)$
Left / right-hand limit	LHL = $\lim_{x \rightarrow c^-} f(x)$ RHL = $\lim_{x \rightarrow c^+} f(x)$
Disproving a multivariable limit	Find two paths that give two different values
Proving a multivariable limit	Squeeze Theorem, or polar-coordinate substitution
Derivative (difference quotient)	$f'(x) = \lim$ as $h \rightarrow 0$ of $[f(x+h) - f(x)] / h$
Partial derivative (preview)	Rate of change in one variable, holding the others fixed

Notation Glossary

Symbol	Meaning
$\lim$	“the limit of” — the value being approached, not necessarily reached
$\rightarrow$	“approaches”, e.g. $x \rightarrow c$ means x gets arbitrarily close to c
$(x, y) \rightarrow (a, b)$	the point (x, y) approaches (a, b) from any direction in the plane
LHL / RHL	left-hand limit / right-hand limit
$f(c)$	the functional value of f at $x = c$
$f'(x)$	the derivative of f with respect to x
h	a small increment added to x , used in the difference quotient

Preparing for Next Lecture

Before the next class, make sure you are comfortable evaluating one- and two-sided limits by direct substitution, and that you can explain in your own words why checking a single path is never sufficient to prove a multivariable limit exists. Next lecture builds directly on the difference-quotient definition of the derivative introduced here, extending it to **partial derivatives** — differentiating with respect to one variable while every other variable is held constant — followed by the **gradient** and **directional derivatives**.